# Smart University Academic & Examination Management System

A university is developing a Smart Academic and Examination Management System (SAEMS) that allows students to manage their academic activities through a single portal. A Student can log in to the system, search available courses, view course details, register for courses, drop registered courses, view their timetable, and check examination schedules. Before a student can register for a course, the system verifies the student's identity and checks whether the student has satisfied the required prerequisites and whether seats are available in the selected course. If a course is full, the student may request to be placed on a waiting list. If the student has outstanding university dues, the system asks the student to clear the dues before completing registration. The student can pay the dues through an external Bank Payment Gateway, which sends the payment result back to SAEMS. After successful registration, the system generates a registration confirmation and updates the student's timetable. A Course Instructor can log in, view the courses assigned to them, view the list of registered students, upload lecture materials, and submit attendance records. The instructor can also enter internal assessment marks, but before marks are submitted, the system validates that all required assessment components have been entered. If an instructor attempts to submit incomplete marks, the system displays an error and allows the instructor to correct them. A Department Coordinator manages course offerings, assigns instructors to courses, approves special course registrations, and reviews requests from students who want to register for courses despite missing prerequisites. The coordinator may approve or reject such requests. An Examination Officer creates examination schedules, assigns examination rooms, publishes the final examination timetable, and records examination results. Before publishing a timetable, the system checks for room and time conflicts. If a conflict is detected, the officer must modify the schedule before publication. A System Administrator manages user accounts, roles, permissions, and system configuration. The administrator can also generate system reports. The university's Identity Authentication Service is used whenever a user logs in, while the Bank Payment Gateway is used whenever a student makes an online payment. The system automatically sends email or SMS notifications to students and instructors after important events such as successful course registration, registration rejection, timetable publication, examination schedule changes, or release of examination results. In exceptional situations, such as a student forgetting their password, the student can initiate password recovery through the authentication service. Similarly, if a student attempts to drop a course after the university's deadline, the system does not normally allow the operation but provides an option to submit a special drop request to the Department Coordinator for approval. Different types of users have different permissions, and some users may perform more than one role within the university.

## Student task

Ask your students to identify and model the system without being given the relationships explicitly. They should determine:

- System boundary
- Primary actors
- Secondary actors
- Human vs. external-system actors

- Candidate use cases
- Actor–use-case associations
- `<<include>>` relationships
- `<<extend>>` relationships
- Actor generalization, if applicable
- Appropriate use-case names
- Any relationships that should not be modeled
- At least one fully dressed use case from their diagram

Important challenge: Tell students that *not every sentence in the scenario automatically becomes a use case*. They must determine which behaviors represent meaningful user goals and which are merely internal system activities.